



Data Engineering

CAPSTONE PROJECT



Global E-Commerce Analytics Platform



Capstone Project

Build an End-to-End Modern Data Engineering Pipeline

Project Title

Global E-Commerce Analytics Platform

**Build a Modern End-to-End Data Pipeline using API → Snowflake → dbt
→ Airflow → Power BI → GitHub Actions**

Project Overview

In this capstone project, students will build a complete modern data engineering solution using industry-standard tools.

The pipeline will automatically collect data from a public REST API, store raw files, load them into Snowflake, transform the data into a data warehouse using dbt, orchestrate the workflow using Apache Airflow, visualize insights in Power BI, and automate testing and deployment with GitHub Actions.

This project simulates a real-world data engineering workflow used in modern organizations.

Learning Outcomes

By completing this project, students will learn how to:

- ✓ Consume REST APIs
- ✓ Work with JSON data
- ✓ Build ETL/ELT pipelines
- ✓ Design Snowflake databases
- ✓ Transform data using dbt
- ✓ Build Star Schemas

- ✓ Create Fact and Dimension tables
- ✓ Schedule pipelines with Airflow
- ✓ Build Power BI dashboards
- ✓ Implement CI/CD using GitHub Actions

Public API

Use the following API:

DummyJSON API

<https://dummyjson.com>

This API contains many endpoints.

Available Endpoints

 Users

`https://dummyjson.com/users`

 Products

`https://dummyjson.com/products`

 Carts

`https://dummyjson.com/carts`

 Posts

`https://dummyjson.com/posts`

 Comments

`https://dummyjson.com/comments`

 Quotes

`https://dummyjson.com/quotes`

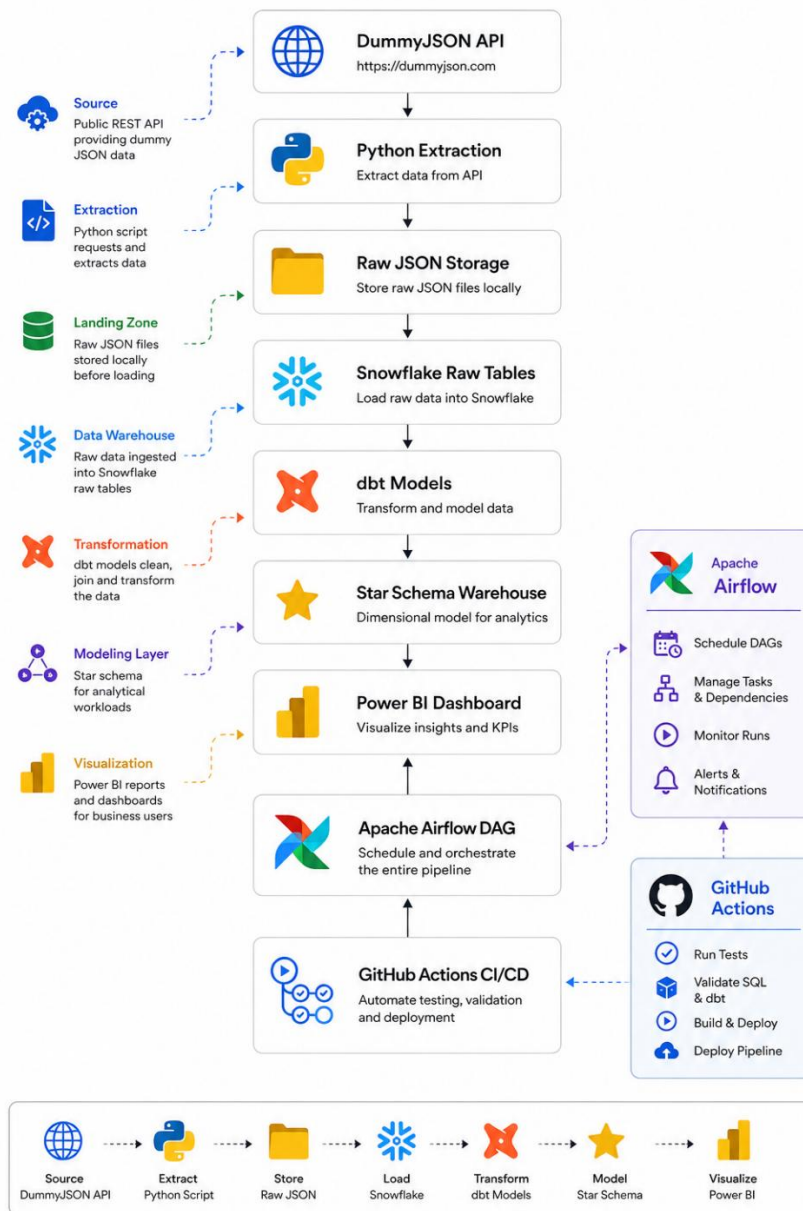
 Recipes

`https://dummyjson.com/recipes`

 Categories

`https://dummyjson.com/products/categories`

Project Architecture



Suggested Project Structure

Capstone_Project/

```

|
| data/
|   raw/

```

```
├── processed/
├── extract/
│   ├── users.py
│   ├── carts.py
│   ├── products.py
│   ├── posts.py
│   └── recipes.py
├── snowflake/
│   ├── load_users.py
│   └── load_products.py
├── dbt_project/
├── airflow/
│   └── dags/
├── powerbi/
├── github/
├── requirements.txt
└── README.md
```

🧩 Step 1 — Extract Data from API

🎯 Objective

Download data from every endpoint.

Tasks

- ✅ Connect to the API
- ✅ Read JSON
- ✅ Handle pagination (if available)
- ✅ Save JSON locally

Example Output

```
raw/  
users.json  
products.json  
carts.json  
posts.json  
comments.json  
recipes.json  
quotes.json
```

Deliverables

- ✓ Python scripts
- ✓ Raw JSON files

Step 2 — Store Raw Data

Objective

Store every API response exactly as received.

Do NOT clean the data.

This is the Data Lake layer.

Folder

```
raw/  
users/  
products/  
carts/  
posts/  
comments/
```

```
recipes/
```

Naming Convention

```
users_2026_07_01.json
```

```
products_2026_07_01.json
```

Deliverables

✓ Timestamped JSON files

Step 3 — Load into Snowflake

Objective

Create staging tables.

Create Database

```
CAPSTONE_DB
```

Schemas

```
RAW
```

```
STAGING
```

```
MART
```

Raw Tables

```
RAW_USERS
```

```
RAW_PRODUCTS
```

```
RAW_CARTS
```

```
RAW_POSTS
```

```
RAW_COMMENTS
```

RAW_RECIPES

Tasks

- ✓ Create tables
- ✓ Load JSON
- ✓ Flatten nested objects

Step 4 — Transform with dbt

Objective

Clean the data.

Tasks

Remove duplicates
Fix null values
Rename columns
Create surrogate keys
Cast data types
Split nested columns

Example

RAW_USERS

↓

STG_USERS

↓

DIM_USERS

dbt Layers

Sources

↓

Staging

↓

Intermediate

↓

Mart

Deliverables

✓ dbt project

✓ dbt documentation

✓ dbt tests

★ Step 5 — Build Star Schema

Fact Tables

FACT_CARTS

FACT_PRODUCT_SALES

Dimension Tables

DIM_USERS

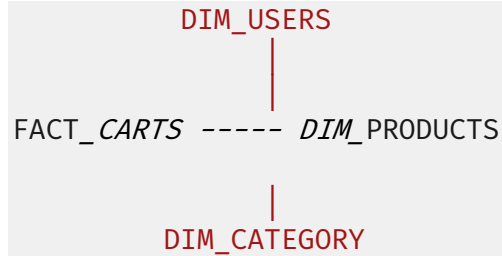
DIM_PRODUCTS

DIM_CATEGORY

DIM_COMPANY

DIM_ADDRESS

Relationship Example



Deliverables

- ✓ Star schema
- ✓ Fact tables
- ✓ Dimension tables

Step 6 — Orchestrate with Airflow

Objective

Automate the entire workflow.

DAG

Extract API

↓

Save Raw Files

↓

Load Snowflake

↓

Run dbt

↓

Data Quality [Check](#)



Refresh Reporting Tables

Tasks

Create DAG

Retry failed tasks

Logging

Email notification (optional)

Scheduling

Deliverables

✓ Airflow DAG

✓ Successful pipeline



Step 7 — Build Reporting Tables

Create analytical tables.

Example

Daily Sales

Monthly Sales

Top Products

Top Categories

Customer Spending

Average **Order Value**

Products **per** Category

Company Distribution

Recipe Difficulty

Posts [per User](#)

Deliverables

✓ Reporting tables

Step 8 — Build Power BI Dashboard

Create an executive dashboard.

Dashboard 1

Sales Overview

Cards

 Total Revenue

 Orders

 Customers

 Products

Charts

 Revenue Trend

 Sales by Category

 Top Products

 Customer Location

Dashboard 2

Customer Analytics

Age Distribution

Gender

Top Buyers

Companies

Dashboard 3

Product Analytics

Category Performance

Brand Performance

Price Analysis

Rating Analysis

Stock Availability

Deliverables

✓ PBIX file

✓ Interactive dashboard

Step 9 — CI/CD using GitHub Actions

Objective

Automatically validate the project.

Trigger

Push

Pull Request

Workflow

Developer Push

↓

GitHub

↓

Install Python

↓

Install Dependencies

↓

Run `Unit` Tests

↓

Run dbt Tests

↓

Lint Python

↓

Success

Deliverables

✓ GitHub Actions workflow

Recommended Technologies

Category	Tool
Programming	Python
API	Requests
Storage	JSON
Database	Snowflake
Warehouse Modeling	dbt
Orchestration	Apache Airflow
Dashboard	Power BI
Version Control	Git
Repository	GitHub
CI/CD	GitHub Actions
IDE	VS Code

By the end of the capstone, students will have built a production-style data engineering solution that demonstrates skills in data ingestion, cloud warehousing, transformation, orchestration, visualization, and CI/CD—closely mirroring the workflow used by professional data engineering teams.